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Define sensors in IoT. Give classification of sensor and explain any four type.

Explain the different types of sensors are used for measuring one of the physical properties and give its representative examples.

Sensors in IoT

Definition:

A sensor is a device that detects physical or environmental conditions such as temperature, light, motion, pressure, etc., and converts them into electrical signals. In IoT systems, sensors collect data from the real world and send it to other devices for processing and decision-making.

Classification of Sensors (with 1-line explanation)

1. Based on Output

- Analog Sensors: Provide continuous output values (e.g., temperature changing slowly).
- Digital Sensors: Provide discrete output (ON/OFF or 0 and 1).

2. Based on Energy Source

- Active Sensors: Require external power to operate.

- Passive Sensors: Work without external power by using energy from the environment.

3. Based on Application

- Sensors are classified based on what they measure, such as temperature, light, motion, pressure, etc.

Explanation of Any Four Types of Sensors (Daily Life Examples)

1. Temperature Sensor

A temperature sensor measures the temperature of surroundings and converts it into an electrical signal.
Daily Example: Used in air conditioners and refrigerators to control temperature automatically.
Application: Weather monitoring, cooling systems

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2. Pressure Sensor

A pressure sensor measures the pressure of air or liquids.
Daily Example: Used in car tires to indicate if air pressure is low.
Application: Automobiles, water supply systems

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3. Motion Sensor

A motion sensor detects movement of people or objects.
Daily Example: Automatic lights that turn ON when someone enters a room or corridor.
Application: Security systems, smart homes

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4. Light Sensor

A light sensor measures the intensity of light.
Daily Example: Smartphone brightness automatically adjusts based on surrounding light.
Application: Street lighting, mobile devices

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5. Humidity Sensors

These sensors measure moisture or humidity in the air.
Examples: Weather apps, room humidity controller
Use: Used in agriculture, weather stations, and homes

Difference (8 Points)

Point	Sensor	Actuator
1. Function	Measures physical quantity	Performs action
2. Role	Input device	Output device
3. Conversion	Physical → Electrical signal	Electrical → Physical action
4. Purpose	Data collection	Control and operation
5. Energy Flow	Receives energy from environment	Uses electrical energy to act
6. Example Devices	Temperature sensor, LDR	Motor, LED, relay
7. Use in System	Sends data to controller	Receives command from controller
8. Output Type	Signal/data	Mechanical movement or action

What is the smart object? Give the security and privacy concerns of smart object in IoT.

Smart Object in IoT

Definition:

A smart object is a physical device embedded with sensors, processors, communication modules, and software that enables it to sense the environment, process data, and communicate with other devices over a network. Smart objects are the basic building blocks of IoT systems.

Security and Privacy Concerns of Smart Objects in IoT

1. Data Privacy Issues

Smart objects collect personal and sensitive data (like location, health, usage patterns). If not protected, this data can be misused.

2. Unauthorized Access

Weak authentication can allow attackers to access and control smart devices without permission.

3. Data Theft (Eavesdropping)

Data transmitted over networks can be intercepted if proper encryption is not used.

4. Data Tampering (Integrity Issues)

Attackers can modify the data being transmitted, leading to incorrect actions or decisions.

5. Device Hijacking

Hackers can take control of smart objects and use them for malicious activities like spying or botnet attacks.

6. Denial of Service (DoS) Attacks

Attackers can overload devices or networks, making them unavailable for normal use.

7. Insecure Software/Firmware

Outdated or poorly designed software can have vulnerabilities that attackers can exploit.

8. Lack of Standard Security Measures

Different manufacturers follow different security practices, leading to inconsistent protection.

9. Physical Security Risks

Devices placed in open areas can be physically accessed, damaged, or tampered with.

10. Identity Spoofing

Attackers can pretend to be a genuine device and gain access to the system.

Explain characteristics of smart object Give the trend in smart object.

Characteristics of Smart Objects

A smart object is an intelligent device in IoT that can sense, process, and communicate data. It has the following characteristics:

1. Unique Identity

Each smart object has a unique ID (like IP address or MAC address) which helps in identification in a network.

2. Sensing Capability

It can sense physical parameters such as temperature, light, motion, etc., using sensors.

3. Processing Capability

It has a microcontroller or processor to process the collected data.

4. Communication Capability

It can communicate with other devices using technologies like Wi-Fi, Bluetooth, Zigbee, etc.

5. Actuation Capability

It can perform actions based on commands, such as switching ON/OFF devices.

6. Energy Efficiency

Smart objects consume less power and often use battery or energy-saving techniques.

7. Intelligence

They can make decisions automatically using predefined logic or AI.

Trends in Smart Objects

1. Miniaturization

Smart objects are becoming smaller, compact, and easy to carry.

2. Increased Intelligence

Integration of Artificial Intelligence and Machine Learning is making devices smarter and more autonomous.

3. Energy Harvesting

Devices are using renewable energy sources like solar or vibration, reducing battery dependency.

4. Improved Connectivity

Advanced technologies like 5G, LoRa, and NB-IoT provide faster and more reliable communication.

5. Edge Computing

Data processing is done near the device, reducing delay and improving performance.

6. Enhanced Security

Improved encryption and authentication techniques are being used to protect data.

7. Interoperability and Standardization

Devices from different manufacturers are becoming compatible through common standards.

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